

# Zakaria Sadeh

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## ABOUT ME

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AI Engineer and Data Scientist with hands-on experience building end-to-end AI systems, from RAG pipelines and multilingual LLM workflows to computer vision and NLP models deployed across media, telecom, and government projects. Skilled in taking models from research to production using Docker, AWS, and modern MLOps practices, with experience leading cross-functional teams as Vice President of a 30+ member Data Science Club. Passionate about pushing the boundaries of AI in the healthcare and education sectors, and always open to connecting with like-minded builders in the field.

## WORK EXPERIENCE

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### AI ENGINEER & DIGITAL TRANSFORMATION INTERN – CIP (Case in Point) – April 2026 - July 2026 – AMMAN, JORDAN

I drove company-wide AI adoption by developing a strategic roadmap for Claude-based agentic workflows and leading AI awareness training. Technically, I built a versatile, end-to-end RAG system using Gemini and ChromaDB for document interaction, and researched Cloudflare and Microsoft infrastructures to enable shadow AI detection. Additionally, I supported business growth by authoring two complete RFP proposals for Qatar-based clients.

### AI ENGINEER & AUTOMATION SPECIALIST – ALJAZEERA – 01/10/2025 – 30/06/2026 – AMMAN, JORDAN

Built a RAG pipeline (BAAI/bge-m3 + ChromaDB) to search large-scale mp3, mp4, PDF, Excel, and Word content, improving retrieval accuracy and discoverability. Built an end-to-end Gemini 2.0 Flash workflow that auto-generates expert-level Arabic/English articles from interview transcripts. Benchmarked Whisper Mid, Whisper Large v3, and Faster Whisper to select the most cost-effective, accurate transcription model for interviews. Containerized applications with Docker and tested AWS deployment, improving portability and reducing deployment friction. Managed weekly workflows and progress tracking for the AI team, ensuring on-time milestone delivery.

### AI ENGINEER (APPRENTICESHIP) – DEVOTEAM – 01/11/2024 – 01/03/2025 – AMMAN, JORDAN

Built deep learning models (EfficientNet, MobileNet, BiT, VGG16, YOLOv8, YOLOv11) for sand-type classification, detection, and segmentation (team of 2). Built ML/DL models (logistic regression, decision tree, KNN, XGBoost, neural networks) to predict Zain customer churn (team of 3). Co-designed an AI solution for Saudi Arabia's Ministry of Human Resources and Social Development on social security beneficiary status management (team of 5). Authored technical/business documentation covering methodology, impact, and model performance for each project.

### AI ENGINEER (APPRENTICESHIP) – TOURSTIFY – 01/06/2024 – 01/10/2024 – AMMAN, JORDAN

Built a YOLO/Roboflow computer vision model to detect gender, age, and weight, with a FastAPI-based data pipeline (team of 2).

## PROJECTS

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### Generative AI & Enterprise LLM Solutions

Engineered end-to-end enterprise GenAI systems for bilingual (Arabic/English) call summarization and automated contract analysis. Fine-tuned LLMs (mT5, DistilBERT) and integrated Google Gemini for clause extraction, risk assessment, and multilingual summarization, with production-ready backend on FastAPI, MongoDB, Redis, Docker, Kubernetes, CI/CD, and JWT authentication.

### Natural Language Processing & Text Analytics

Built NLP models for sentiment analysis, topic classification, and large-scale text mining on call-center transcripts and social media, using DistilBERT, BERT, AraBERT, GPT-2, LSTM/RNN, Word2Vec, FastText, and tf-idf embeddings

### Computer Vision for Public Safety & Smart Infrastructure

Developed YOLO-based object detection and tracking for smart-city infrastructure reporting, crowd management, and school-transport safety (Face ID recognition, route optimization), plus satellite-imagery models for vacant-land and residential-unit detection.

### Business Intelligence & Economic Forecasting

Built Power BI dashboards to visualize and forecast key economic indicators and sector-level growth trends.

### Applied Research & Predictive Modeling

Developed an RNN-based model for real-time monitoring and forecasting of geomagnetic storm activity to support time-sensitive research decisions.

### Kaleem platform (client project)

I engineered an end-to-end AI transcription pipeline by fine-tuning Whisper tiny on a diverse dataset of 300 child and adult audio recordings, achieving a 20x accuracy improvement over the base model. To operationalize this, I developed a FastAPI backend that seamlessly receives audio input, processes it through the fine-tuned model, and serves the transcribed text via API.

## EDUCATION AND TRAINING

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01/10/2021 – 24/02/2026

Amman, Jordan

**BACHELOR OF SCIENCE IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (HNC & HND)** Al-Hussein Technical University (HTU)

- programming and software development (Python, data structures & algorithms, object-oriented and procedural programming)
- mathematics and statistics for computing (functional math, discrete math, applied analytical models)
- data science and analytics (data analytics, data mining, big data analytics and visualization, data engineering)
- machine learning and deep learning (supervised/unsupervised learning, neural networks, model deployment/MLOps)
- artificial intelligence and intelligent systems (AI foundations, natural language processing, computer vision, pattern recognition)
- database design and development, and business intelligence for data-driven decision-making
- computer networking, operating systems, and information security fundamentals
- professional and research skills (project planning, computing research project, capstone project, industry training)

**MACHINE LEARNING SPECIALIZATION** Stanford University 01/10/2023 – 01/02/2024  
Completed a Machine Learning Specialization, mastering key concepts including supervised and unsupervised learning, recommender systems, and reinforcement learning. Applied best practices in model building, gaining hands-on experience to solve real world problems

**DATA MINING SPECIALIZATION** University of Illinois Urbana-Champaign 01/09/2024 – 15/01/2025  
This program enhanced my understanding of data mining techniques for both structured and unstructured data, covering key topics such as pattern discovery, clustering, text retrieval, text mining, analytics, and data visualization.

**TENSORFLOW DEVELOPER PROFESSIONAL CERTIFICATE** DeepLearning.AI 05/01/2023 – 14/03/2023  
Mastered TensorFlow for computer vision and natural language processing, implementing overfitting prevention strategies. Trained neural networks using RNNs, GRUs, and LSTMs on real-world data to enhance model performance.

**PYTHON FOR DATA SCIENCE, AI & DEVELOPMENT** IBM 02/02/2023 – 21/03/2023  
- Python programming fundamentals: syntax, data types, expressions, variables, and string operations  
- programming logic using data structures, conditionals, loops, functions, exception handling, and object-oriented programming (classes and objects)  
- data analysis with Python libraries (Pandas, NumPy) using Jupyter Notebooks  
- web-based data access: REST APIs (via the requests library) and web scraping with BeautifulSoup

**INTRODUCTION TO MODEL CONTEXT PROTOCOL** Claude Education 08/2026  
Focused on building fully functional MCP servers and clients to enable standardized, seamless communication between AI models and external data systems without tedious integration code.

**INTRODUCTION TO AGENT SKILLS** Claude Education 08/2026  
Covered creating, configuring, and sharing reusable markdown instructions that allow Claude Code to automatically apply specific, context-efficient workflows to matching tasks.

**CLAUDE CODE IN ACTION** Claude Education 08/2026  
Mastered terminal-based agentic coding workflows, enabling Claude to autonomously navigate repositories, execute commands, and perform multi-step file refactoring directly from the CLI.

**CLAUDE WITH THE ANTHROPIC API** Claude Education 08/2026  
Provided comprehensive training on building AI-powered applications, covering advanced prompt engineering, custom tool calling, Retrieval-Augmented Generation (RAG) pipelines, and complex agent architectures.

**SKILLS**

**Soft Skills**

Team Collaboration Leadership Communication Problem-Solving Project Management Teamwork Work Ethic Creativity Time Management Adaptability Attention to Detail Critical Thinking public speaking

**Technical Skills**

Frameworks and Libraries: FastAPI, Open CV, Tensorflow, Keras, Pandas, Seaborn, Streamlit Word Embedding MongoDB RAG pipeline Shadow AI computer vision Python Deployment Python (computer programming) natural language processing RESTful API design Java (computer programming) Database design and development LangChain Python - Numpy, Pandas Backend API development Docker data engineering business intelligence Python (NumPy, Pandas, NLTK, Matplotlib, sklearn, TensorFlow) Artificial Intelligence and Machine learning Computer programming (C/C++, C#, Python) Machine Learning ( AI, DL, CNN) Git/Github, Docker, Gitlab AWS deployment Large Language Model integration Cloud concepts Chroman DataBase Word Vector DataBase Streamlit Microsoft Office agents (agentic pipeline) LoRA fine-tuning clean, testable code Scikit-learn Backend Claude Certified Claude MCP

**LANGUAGE SKILLS**

Mother tongue(s): **ARABIC**  
Other language(s): **PROFESSIONAL ENGLISH** (Cambridge certified)